KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 28 · SCIENCE

Heat and Thermal Energy

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: What Do I Already Know?

Answer with what you know right now. It is OK to guess!

1. **Review (Week 27):** About 2.5 percent of Earth's water is fresh. Where is most of that fresh water stored? Name the two biggest places.

2. Predict: You hold an ice cube and your hand gets cold. What do you think is moving, and in which direction?

3. A metal spoon and a wooden spoon sit in the same bowl of hot soup. Which handle do you think gets hot first? Why?

4. Name two ways your family keeps food or drinks hot or cold. What do you think makes them work?

5. Draw how you think the Sun's warmth reaches Earth across space. Label your drawing.

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POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Following the Heat

Use evidence from today's lesson. Write in complete sentences.

1. **Sort it.** Name the way heat moves in each: (a) a pot handle heating on the stove, (b) warm air rising from a heater vent, (c) your face warming in the sunshine.

2. **Two-step.** Change 15 degrees Celsius into Fahrenheit (multiply by 1.8, then add 32). Show both steps.

3. **Find the mistake.** Pieter says, "My mittens make heat for my hands." Explain what is wrong, using the word *insulator*.

4. **Data.** Pinar's bare mug cooled from 120 to 96 degrees in 30 minutes, and her foam-wrapped mug cooled from 120 to 108. How many degrees did each cool? Which one insulated better?

★ **Challenge: Explain.** A vacuum bottle keeps soup hot for hours. How does it slow conduction, convection and radiation?

Looking Ahead: Next week: People Protecting Earth. You will use real data to see how people change land, water, air and living things, and meet the scientists and communities fixing the damage.